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Section: B2

*exp10.y

```
%{
#include "y.tab.h"
%}

%%

"for"        { return FOR; }
"int"        { return INT; }
"<="|">="|"=="|"!="|"<"|">"   { return RELOP; }
[a-zA-Z]+    { return ID; }
[0-9]+       { return NUM; }
[ \t\n]      ;
.            { return yytext[0]; }

%%

int yywrap(void)
{
    return 1;
}
```

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```
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~$ yacc -d prac10.y
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~$ flex prac10.l
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~$ gcc y.tab.c lex.yy.c -o prac10 -ll -lm
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~$ ./prac10
Enter a for loop statement:
for (i = 0; i <= 8; i++);
Valid for loop
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~$
```

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*exp10.y

```
%{
#include <stdio.h>

int yylex(void);
int yyerror(const char *s);
%}

%token FOR INT ID NUM RELOP

%%

stmt:
    FOR '(' init ';' cond ';' update ')' stmt
    {
        printf("Valid for loop\n");
    }
    | ';'
    ;

init:
    INT ID '=' NUM
    | ID '=' NUM
    ;

cond:
    ID RELOP NUM
    ;

update:
    ID '+' '+'
    | ID '-' '-'
    ;

%%

int main(void)
{
    printf("Enter a for loop statement:\n");
    yyparse();
    return 0;
}

int yyerror(const char *s)
{
    printf("Invalid for loop\n");
    return 0;
}
```